

Teaching line, not a question slide.

两小时考前冲刺，一对一，投屏远距离。学生基础好，节奏可以快。

时间表：Lesson 1 Green Machines 0:00–0:50；Lesson 2 Metallic materials 0:50–1:40；Mini mock 1:40–1:57；作业 1:57–2:00。

开课前发纸：Classwork (Part A–F) 和 Mini mock 先发 Classwork，Mini mock 到 1:40 再发；Homework 两份下课时发。全部下课收回。

评分语言统一用 A / M / E：A 写出事实，M 把原因连上，E 把整条链写完整并用数据或对比。

课件页脚的页码指她的两本 booklet：Green Machines p.X 和 Metals p.X。

Teaching line, not a question slide.

时间 0:00-0:02。

告诉她这一小时的三段：Loop 1 reproduction, Loop 2 photosynthesis and respiration, Loop 3 transport。每段结尾 2-3 道 Practice, 最后 10 分钟做 Classwork Part A-C。

她这本 booklet 基本都写了, 所以今天不是重新教, 而是把答案写到 M 和 E。

Say it:

photosynthesis = foh-toh-SIN-thuh-sis

respiration = res-pih-RAY-shun

Q1 anther

Q2 fertilisation

Q3 water, warmth, oxygen

Q4 carbon dioxide

Q5 xylem

Short solution:

Q1 anther 在 filament 顶端，两者合起来是 stamen

Q2 pollen 从 anther 到 stigma 是 pollination；两个 nucleus 融合才叫 fertilisation

Watch for:

Q2 答 pollination 说明两个词混了，这是她最容易丢的定义分

Q5 答 “veins” 不够，要写 xylem

Diagnostic:

Q2 错：Loop 1 的 “From pollen to seed” 多花 1 分钟

Q4 答 oxygen：把 photosynthesis 和 respiration 的方程式记反了，Loop 2 先讲方程式

Q5 答 phloem：xylem 和 phloem 混淆，Loop 3 的 Quick review 先做

Say it:

pollination = pol-ih-NAY-shun

fertilisation = fer-tih-ly-ZAY-shun

ovule = OV-yool

germinate = JER-mih-nayt

xylem = ZY-lum

phloem = FLOH-um

stamen = STAY-mun

Teaching line, not a question slide.

时间 0:02-0:03。

MAKE food = photosynthesis (叶子) ; MOVE materials = transport: water 走 xylem, glucose 走 phloem; MAKE new plants = reproduction: flower、seed、germination。

问她：“testing a leaf for starch 属于哪一个？” 答：MAKE food，因为 starch 是储存起来的 glucose。

考试时先判断是哪一件事，再用那一块的术语，答案就不会跑题。

1. A stigma, B style, C ovary, D anther, E filament, F sepal
2. anther and filament (D and E)
3. stigma, style and ovary (A, B and C)
4. the ovary (C)

Short solution:

4. fertilisation 之后 ovary wall 变成 fruit, 里面的 ovule 变成 seed

Watch for:

1. sepal 和 petal 不要混: sepal 在花的最外面最下面, 花还是花苞时保护它
3. carpel 也叫 pistil, 两个名字都可以

Say it:

sepal = SEE-pul

carpel = KAR-pul

Teaching line, not a question slide.

时间 0:05-0:07。

空格答案：Asexual 一栏 one parent / no gametes / identical (clones); Sexual 一栏 offspring show variation / flowers and seeds。

问题答案：asexual reproduction, 因为只有 one parent, 没有 gametes, plantlets 和 parent plant identical (clones)。

她在 booklet p.4 把 “This specific process involves gametes” 也打了勾, 这是 L2: asexual 一定没有 gametes。

Say it:

asexual = ay-SEK-shoo-ul

gametes = GAM-eets

Teaching line, not a question slide.

时间 0:07-0:08。

空格：different (show variation) / survive the change (resist the disease) / reproduce / continues。

问题答案：the plants are clones with the same genetic material，所以如果一株会得这种病，所有植株都会得，没有一株有抵抗力。

这是 Explain 长答题的标准链条。她 booklet p.7 写 “helps the species to live”，只到 A。要拿 E 必须写出 “some survive → reproduce → species continues”，并且和 asexual 对比。

Teaching line, not a question slide.

时间 0:08-0:10。

顺序：F, B, A, E, D, C。

Pollination = F 和 B (pollen 从 anther 到 stigma) 。 Fertilisation = D (两个 nucleus 融合，形成 zygote) 。

提醒 L1：pollen 落在 stigma 上还不是 fertilisation。

D 之后 zygote 长成 embryo，在 seed 里面。

Say it:

micropyle = MY-kroh-pile

zygote = ZY-goht

embryo = EM-bree-oh

Teaching line, not a question slide.

时间 0:10-0:12。

空格：large, brightly coloured, scented, with nectar / large and feathery, hanging outside / firmly attached inside the flower / small, light, made in huge amounts。

问题答案：so the wind can blow the pollen away from the anthers。只写“they hang outside”是 A；写出“so the wind can carry the pollen”才是 M。

这是 L3：每个 feature 后面都要加“so that ...”。

1. runners have no gametes: the new plant is identical to the parent, a clone (L2)
2. pollen is carried from the anther to the stigma, not the other way (L1)
3. that transfer is pollination, not fertilisation (L1)

Short solution:

Runners are asexual: one parent, no gametes, so the new plants are identical to the parent. Flowers: pollen is carried from the anther to the stigma (pollination); the pollen nucleus fuses with the ovule nucleus (fertilisation). The seeds show variation because two parents are involved.

Watch for:

最后一句是对的。让她先判断每一句，而不是看到 Sam 就全部改掉。

1. P testa, Q plumule, R radicle, S cotyledon
2. water, warmth, oxygen
3. so the new plant does not compete with the parent for light, water, minerals and space

Short solution:

1. testa 是保护种子的硬外皮, water 让它变软; radicle 向下长成 root, 固定植物并吸水; plumule 向上长成 shoot; cotyledon 是 food store, 在叶子能 photosynthesise 之前提供能量
2. oxygen 用于 respiration, 释放生长需要的能量

Watch for:

1. booklet 印的是 “Radical”, 考试要拼 radicle
3. 只写 “to spread out” 是 A; 要写出 competition 和资源名称

Say it:

plumule = PLOO-myool

radicle = RAD-ih-kul

cotyledon = kot-ih-LEE-dun

dispersed = dis-PERST

Q1 insect-pollinated: large white petals and scent attract insects; nectar is food for insects

Q2 cuttings are asexual, so the new plants are identical to the best bush and keep its features; seeds come from sexual reproduction and show variation

Q3 only B germinates; D has no oxygen, so the seed cannot respire to release energy for growth

Short solution:

Q3 A 没有 water; C 没有 warmth (fridge 里温度太低) ; D: boiling 把水里溶解的 oxygen 赶走, oil 层不让空气重新溶进去

Watch for:

Q1 白色花瓣也算 “large, showy”, 不一定要彩色

Q3 C 的标准答案写 “too cold / no warmth”。D 一定要写 oxygen 的用途 (respiration 释放能量), 否则只有 M

Q3 这个 boiled water + oil 的设计和 Lesson 2 的 rusty nail 完全一样, 等一下会用到

1. A waxy cuticle, B upper epidermis, C palisade layer, D spongy layer, E vein (xylem and phloem), F lower epidermis, G guard cell, H stoma

2. the palisade layer (C)

Short solution:

2. palisade cells 又高又窄、排得很紧，在叶子上部，含最多 chloroplasts

Watch for:

1. H 的复数是 stomata；拼写常错

1. 她 booklet p.15 把 “transports water and minerals” 填成 spongy layer。应该是 vein 里面的 xylem (L7)

Say it:

cuticle = KYOO-tih-kul

epidermis = ep-ih-DER-mis

palisade = pal-ih-SAYD

stomata = STOH-muh-tuh

chloroplasts = KLOR-uh-plasts

Teaching line, not a question slide.

时间 0:19-0:21。

方程式: carbon dioxide + water $\rightarrow$ glucose + oxygen; 箭头上写 light energy, 下面写 chlorophyll。

carbon dioxide 从 stomata 进入; water 从 root hairs 进入 (osmosis), 经 xylem 到叶子。

chlorophyll: absorbs light energy (真实的科学)。booklet p.20 写的是 chlorophyll “speeds up the reaction”, 按学校的说法两种都可以接受。

L4: light 和 chlorophyll 不是反应物, 写在箭头上下。

Say it:

chlorophyll = KLOR-uh-fil

osmosis = oz-MOH-sis

Teaching line, not a question slide.

时间 0:21-0:23。

方程式: $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} (+ \text{energy})$ 。在 mitochondria 里进行。

表格: Day 两个都 active; Night photosynthesis not active, respiration active。

问题答案: respiration。

L5: 植物白天也 respire, 只是白天 photosynthesis 更快, 所以白天净放出 oxygen。

Say it:

mitochondria = my-toh-KON-dree-uh

Teaching line, not a question slide.

时间 0:23-0:25。

1. kills the leaf and stops its reactions (so the starch stays)
2. removes the chlorophyll, so the colour change can be seen
3. softens the leaf and washes off the ethanol
4. starch is present; 没有 starch 时保持 brown / yellow-brown

问题答案: ethanol is flammable, 不能靠近明火。

实验题常考 “why each step”, 她最容易只写步骤不写原因。

Say it:

ethanol = ETH-uh-nol

iodine = EYE-uh-dyne

(a) J (green, not covered): blue-black

(b) K (white): stays brown

(c) L (under the foil): stays brown, even where it is green

Short solution:

J 有 chlorophyll 也有 light, 进行 photosynthesis, glucose 变成 starch 储存

K 没有 chlorophyll, 不能 photosynthesise, 没有 starch

L 没有 light, 不能 photosynthesise, 没有 starch

Watch for:

这是 booklet p.22 第 2 题, 她没写。每个区域都要说“缺什么”: K 缺 chlorophyll, L 缺 light

Say it:

variegated = VAIR-ee-uh-gay-tid

1. carbon dioxide + water → glucose + oxygen: Sam swapped oxygen and carbon dioxide (L4)
2. blue-black shows starch, not glucose (L6)
3. plants respire all the time, day and night (L5)

Short solution:

carbon dioxide + water → glucose + oxygen, with light energy above the arrow and chlorophyll below. Blue-black with iodine shows the leaf contains starch. Plants respire all the time; they photosynthesise only in the light.

Watch for:

实验步骤那一句是对的。

Q1 plant 1 stays brown (no starch); plant 2 turns blue-black (starch)

Q2 at night only respiration happens, which gives out carbon dioxide; in the day photosynthesis takes in carbon dioxide faster than respiration gives it out

Short solution:

Q1 jar 1 没有 carbon dioxide, 不能 photosynthesise, 不能做 glucose, 所以没有 starch; jar 2 有 carbon dioxide, 是 control

Q2 要写清两个过程, 以及白天 “photosynthesis is faster than respiration”

Watch for:

Q1 她 booklet p.21 只写了 “there won't be photosynthesis”, 没写原因 (缺 carbon dioxide), 只能拿 A

Q2 写 “plants respire at night” 只答了一半; 只写 “plants breathe out carbon dioxide” 不给分

1. D
2. A
3. C
4. B
5. E

Short solution:

记法: xylem 运 water 向上; phloem 运 food (glucose) 到需要或储存的地方

Watch for:

2. phloem 的拼写: ph 开头, -oem 结尾

5. booklet p.16 说 guard cells “swell to close” stomata。真实情况是 guard cells 吸水 swell 时 stomata 打开。考试如果问 swell 和 open 的关系, 按 booklet 写, 并提醒她两种说法都见过

Teaching line, not a question slide.

时间 0:31-0:33。

空格：osmosis / xylem / stomata。

问题答案：transpiration。

Say it:

transpiration = tran-spih-RAY-shun

Teaching line, not a question slide.

时间 0:33-0:35。

空格: evaporates (is lost) / pulls / xylem / leaves / xylem vessels。

她 booklet p.17 的答案是 “The water in the cylinder are being transport up to the stem. The food colour will change the colour of plant”，只到 A: 没有 xylem，没有原因。

L9: 不能写 “the plant sucks / drinks”，要写 transpiration 造成 pull。

Teaching line, not a question slide.

时间 0:35-0:37。

1. the small circles (六边形是 sugar molecules)
2. from right to left, into the sugar solution
3. the sugar molecules are too large to fit through the membrane

定义空格: water / high / low / semi-permeable。

L8: 说 “sugar moves” 或者漏掉 membrane 都会丢分; “high to low” 一定要说是 water 的浓度。

Say it:

semi-permeable = sem-ee-PER-mee-uh-bul

molecule = MOL-ih-kyool

1. Z (the Visking tubing) is the root hair; X (the capillary tube) is the stem
2. water moves into the sucrose solution by osmosis, from the beaker (high concentration of water) through the membrane, so the liquid is pushed up the tube
3. it has a large surface area, so it can absorb more water by osmosis

Short solution:

2. Visking tubing 是 semi-permeable membrane: water 可以过, sucrose 不能过

Watch for:

1. 这是 booklet p.18 的 Conclusions, 她没写
3. 只写 “absorbs water” 不够, 要说 surface area

Say it:

sucrose = SOO-krohs

1. the water goes up the xylem, not the phloem (L7)
2. not “sucks because it is thirsty” : water evaporates through the stomata (transpiration) and pulls water up the xylem (L9)
3. osmosis is from a high to a low concentration of water (L8)

Short solution:

Water evaporates from the flower and leaves through the stomata (transpiration). This pulls water, with the dye, up the xylem, so the petals turn blue. Water enters root hairs by osmosis, from a high to a low concentration of water.

Watch for:

第 3 句只算一个 error (方向反了)。

Q1 osmosis; xylem

Q2 (a) 1.8 mm per minute

Q2 (b) water enters the sugar solution by osmosis through the membrane, so the volume rises

Q3 the salty water has a lower concentration of water than the root cells, so water moves out of the roots by osmosis; the plant loses water and wilts

Short solution:

Q2 (a) rise = $48 - 12 = 36$ mm; rate = $\frac{36 \text{ mm}}{20 \text{ min}} = 1.8$ mm per minute

Q3 先比较 water concentration (外面低、里面高)，再说方向 (out of the roots)，最后说结果 (wilts)

Watch for:

Q2 (a) 常见错误：用 48 除以 20 = 2.4，忘了减去开始的 12 mm

Q3 这是 osmosis 换个角度问：水从 high 到 low，所以这次是从根里往外走

Teaching line, not a question slide.

时间 0:38-0:39。

让她每一条说一个今天课上出现过的例子：L1 Spot the error 1；L2 spider plant；L4 和 L6 Spot the error 2；L7 carnation。
L8 和 L9 在作业 hw1 里还会再练。

Teaching line, not a question slide.

时间 0:39-0:40。

让她闭上眼说一遍这四句，再看屏幕对一下。

Teaching line, not a question slide.

时间 0:40-0:50。

她做的时候老师不要讲，最后 2 分钟对 Part C 的图 (标尺、轴标签、单位、描点)。

答案在 classwork_answers.pdf: Part A 重点是 pollination 和 fertilisation 的区分; Part B 重点是 bell jar 实验的 control; Part C 的图从 0 到 25 分钟是直线上升，之后变平。

时间不够就把 Part C 的画图留到下课后。

Teaching line, not a question slide.

时间 0:50-0:51。

这本 booklet 从 p.9 Alloys 往后她几乎没写，学校教过。所以 Loop 2-4 每一页都要让她自己说，不要老师先讲。

四段：Loop 1 properties and density, Loop 2 heat treatment and alloys, Loop 3 corrosion, Loop 4 reactions and the reactivity series。

Say it:

reactivity = ree-ak-TIV-ih-tee

alloys = AL-oyz

corrosion = kuh-ROH-zhun

Q1 any two: lustrous, malleable, ductile, sonorous, good conductors of heat and electricity, high melting point

Q2 iron, nickel, cobalt

Q3 density = $\frac{\text{mass}}{\text{volume}}$

Q4 liquid

Q5 annealing

Short solution:

Q4 温度高于 melting point 就是 liquid

Q5 annealing = cool slowly; quenching = cool quickly in cold water

Watch for:

Q1 写 “shiny” 可以，但考试更喜欢 lustrous；写 “strong” 不算通用性质

Q3 写成 mass $\times$ volume 是 booklet p.4 她表格里出过的错 (L10)

Diagnostic:

Q3 错：Loop 1 的 density 两页要慢讲

Q5 错：heat treatment 名称不熟，Loop 2 的 Quick review 先做

Say it:

malleable = MAL-ee-uh-bul

ductile = DUK-tile

lustrous = LUS-trus

annealing = uh-NEE-ling

gallium = GAL-ee-um

sonorous = SON-uh-rus

quenching = KWEN-ching

Teaching line, not a question slide.

时间 0:54-0:55。

上排：atoms 怎么排列 → physical properties (malleable, conductor, density) → uses。alloy 也在这一排。

下排：reactivity → 和 oxygen、water、acid 的反应 → rusting 和怎么保护。

问她：“为什么 gold 做首饰？”两排都用到：lustrous (上排) + very unreactive，不会 corrode (下排)。

1. B
2. D
3. A
4. C
5. E

Short solution:

记法: malleable 像 mallet (锤子) ; ductile 像 duct (管子、拉长)

Watch for:

1. 拼写: 两个 l, -eable 结尾 (malleable)
4. sonorous 只有一个 n 在开头

Teaching line, not a question slide.

时间 0:57-0:59。

(a) W: strength high, melting point high

(b) Y: electrical conductivity very high

(c) X: conductivity high and density low (2.7), 所以 light

(d) X 和 Z: melting point 都低于 800°C

这一页先只说选哪个，下一页再练怎么写理由。

Teaching line, not a question slide.

时间 0:59–1:01。

E 级范例：I choose X because it has a high electrical conductivity and a low density of 2.7 g/cm^3 . A light cable puts less weight on the pylons and does not sag. Y conducts even better, but its density is 8.9 g/cm^3 , more than three times heavier.

她 booklet p.6 第 5 题写 “Copper, good conductivity and won't burn you to death”：有 property，但没有 “为什么这个 property 重要”，也没有对比，只有 A (L12)。

Teaching line, not a question slide.

时间 1:01–1:03。

空格：mass / volume；g/mL；length × width × height；displacement。

问题答案：29 – 22 = 7 mL (读刻度：before 22 mL, after 29 mL)。

1 cm³ = 1 mL，所以 g/cm³ 和 g/mL 是同一个单位。

Say it:

displacement = dis-PLAYS-ment

Teaching line, not a question slide.

时间 1:03–1:04。

Step 1: $33.0 - 30.0 = 3.0 \text{ mL}$

Step 2: $\frac{25.5 \text{ g}}{3.0 \text{ mL}}$

Step 3: 8.5 g/mL

L11: volume 是两次读数的差，不是最后的读数。L10: 除法，单位 g/mL。

1. volume = $25.0 - 20.0 = 5.0$ mL, not 25.0 mL (L11)
2. density = mass divided by volume, not multiplied (L10)
3. the unit is g/mL, not g (L10)

Short solution:

$$\text{volume} = 25.0 - 20.0 = 5.0 \text{ mL}; \text{ density} = \frac{39.5 \text{ g}}{5.0 \text{ mL}} = 7.9 \text{ g/mL (和 booklet p.6 表里的 iron 一样)}$$

Watch for:

她 booklet p.4 最后一行也是乘了 (16.55 g 和 2 cm 得出 33.25)，体积单位还写成 cm。答案 987.5 明显不合理，要养成检查“是否合理”的习惯。

Q1 5.4 g/cm^3

Q2 silver (10.5 g/mL)

Q3 copper is almost as good a conductor as silver, is much cheaper, and is ductile, so it can be drawn into wires

Short solution:

$$\text{Q1 volume} = 2.0 \times 2.0 \times 3.0 = 12.0 \text{ cm}^3; \text{ density} = \frac{64.8 \text{ g}}{12.0 \text{ cm}^3} = 5.4 \text{ g/cm}^3$$

$$\text{Q2 volume} = 43.0 - 40.0 = 3.0 \text{ mL}; \text{ density} = \frac{31.5 \text{ g}}{3.0 \text{ mL}} = 10.5 \text{ g/mL}, \text{ 对表得 silver}$$

Q3 E 要有对比: booklet p.6 的数据 silver 60、copper 50, 差别不大, 但 silver 是贵金属

Watch for:

Q2 先减读数再除, 答案要和表对上才算数

Say it:

aluminium = al-yoo-MIN-ee-um

1. quenching, B
2. annealing, A
3. tempering, C
4. tempering

Short solution:

4. hair clip 要 springy, 所以是 tempering

Watch for:

4. 她 booklet p.8 第 2 题写 “Annealing because it modified easily”, 这是错的: annealing 让钢变软, 夹子会弯了回不去

Say it:

tempering = TEM-per-ing

Teaching line, not a question slide.

时间 1:07-1:08。

答案：the layers slide over each other easily, 所以 pure metal 是 soft 的，也因此 malleable 和 ductile。

booklet p.9: pure iron is too soft for many uses。

Teaching line, not a question slide.

时间 1:08–1:09。

空格: different / disrupts / slide over each other / harder。

L13: 只写 “because it is a mixture” 不给分, 必须写 atoms of different sizes 和 layers can't slide。

Teaching line, not a question slide.

时间 1:09–1:10。

空格: tin / joining electrical wires / cutlery and sinks (resists corrosion) / teeth braces and spectacle frames (shape memory)。

solder: melting point 低, 容易熔化, 冷却后在常温下是 solid。

alloy 是 mixture: 比例可以变, 每种成分保留自己的 chemical identity; compound 是固定比例、化学结合。

nitinol 是 smart alloy (shape memory alloy): 弯了以后加热或通电会回到原来的形状。

Say it:

chromium = KROH-mee-um

nitinol = NIT-ih-nol

titanium = ty-TAY-nee-um

Teaching line, not a question slide.

时间 1:10–1:11。

1. pure X (100 %) 262 °C; pure Y (0 % X) 330 °C

2. 70 % X, about 190 °C

3. the melting point falls steadily as the percentage of X increases

提醒 booklet 的 “A SLurP of Tea”: Axes, Scale, Label, Plot, Title。她 p.10 的 solder 图还没画，留在作业里。

结论 (和 booklet p.11 Q1 同一个思路)：alloy 的 melting point 可以比两种纯金属都低。

Q1 quenching

Q2 zinc atoms are a different size from copper atoms; they disrupt the regular layers, so the layers cannot slide over each other easily

Q3 spring: tempering, so it is tough and springy and returns to shape; door panel: annealing, so it is soft and easy to press into shape

Short solution:

Q3 每一个都要 “treatment + 得到的 property + 为什么这个用途需要它”

Watch for:

Q2 只写 “alloys are harder” 是复述题目，0 分

Q3 门板写 quenching 是错的：太 brittle，一撞就裂

1. true
2. false
3. true
4. false
5. true

Short solution:

2. 只有 iron (和 steel) 会 rust; 其他金属是 corrode
3. booklet p.13: salt solution 和 higher temperature 都会加快 rusting

Watch for:

LO 要求 “distinguish between corrosion and rusting”: rusting 是 corrosion 的一种, 只用于 iron

Teaching line, not a question slide.

时间 1:12–1:14。

A: water (moisture) from the air, 所以 A 是 dry air

B: dissolved oxygen (air); air from dissolving back into the water

C: water and oxygen (air)

这个设计和 Lesson 1 Practice Q3 的种子实验一样: boiled water + oil = 没有 oxygen。

Say it:

calcium = KAL-see-um

Teaching line, not a question slide.

时间 1:14–1:15。

结果：A no rust; B no rust; C rust。

空格：C / water / oxygen / both water and oxygen。

她 booklet p.17 的结果写 B “slight rusting, barely can see”，是错的。B 没有 oxygen，不应该生锈；如果真的有一点，说明 boiled water 又溶进了空气 (L14)。

1. P has air; it does not rust because it has no water (L14)
2. Q does not rust: boiled water has no oxygen and the oil keeps air out (L14)
3. iron needs both water and oxygen to rust (L14)

Short solution:

P: no rust, because the calcium chloride removes the water. Q: no rust, because boiling removes the oxygen and the oil keeps air out. R: lots of rust, because it has water and oxygen. So iron needs both water and oxygen to rust.

Watch for:

R 那一句是对的。steel 也会 rust, 因为 steel 主要是 iron。

Teaching line, not a question slide.

时间 1:16–1:17。

空格：painting；moving parts，paint would rub off；galvanising；made of stainless steel (alloying)。

最后一句：water and oxygen (air)。

booklet p.13：选哪种方法看 object 的大小、用途和成本。

Say it:

galvanising = GAL-vuh-ny-zing

Q1 water and oxygen

Q2 the air near the sea carries salt, and salt solution speeds up rusting

Q3 change: tap water or salt water; measure: how much rust after 3 days; keep the same: type and size of nail (cleaned with sandpaper), volume of water, temperature, time, tubes open to the air; repeat each tube

Short solution:

Q3 E 要有: independent variable、dependent variable、至少三个 controlled variables、重复实验

Watch for:

Q3 她实验题最容易漏 “how to measure”: 写 “compare the amount of rust” 或 “measure the mass of the nail before and after” 都可以

1. metal oxide
2. metal hydroxide + hydrogen
3. metal chloride + hydrogen
4. squeaky pop

Short solution:

2 和 3 都放出 hydrogen gas, 所以都可以用 lit splint 测

Watch for:

2. 写成 metal oxide + hydrogen 是常见错误 (L15)

Say it:

hydroxide = hy-DROK-side

hydrochloric = hy-droh-KLOR-ik

Teaching line, not a question slide.

时间 1:21–1:23。

(a) iron oxide

(b) lithium hydroxide + hydrogen

(c) aluminium + hydrochloric acid $\rightarrow$ aluminium chloride + hydrogen

(d) no reaction: gold 在 reactivity series 的最下面

考试写 word equation: 用词, 不用化学式; 箭头不要写成等号。

Say it:

lithium = LITH-ee-um

Teaching line, not a question slide.

时间 1:23–1:25。

1. yes: tin 在 “react with acids” 的范围里 (反应很慢)

2. gold is so unreactive that it does not react with oxygen or sulfur, so it stays as the pure metal; iron is more reactive, so it is found as compounds (ores) and must be extracted

这一页对应 booklet p.20 Summary 的图，重画成四组。

Say it:

potassium = puh-TAS-ee-um

magnesium = mag-NEE-zee-um

mercury = MER-kyuh-ree

1. calcium + oxygen → calcium oxide, not calcium oxygen (L15)
2. iron + hydrochloric acid → iron chloride + hydrogen, not water (L15)
3. silver + hydrochloric acid → no reaction: silver is below the acids in the series (L16)

Short solution:

calcium + oxygen → calcium oxide; iron + hydrochloric acid → iron chloride + hydrogen; potassium + water → potassium hydroxide + hydrogen (correct); silver + hydrochloric acid → no reaction

Watch for:

第 3 条是对的，error 的编号按找到的顺序 1-3，不按 Sam 的行号。

Q1 aluminium + oxygen → aluminium oxide

Q2 S, P, R, Q

Q3 add same-sized pieces of zinc and copper to the same volume of the same dilute hydrochloric acid; zinc fizzes (hydrogen), copper does not react

Short solution:

Q2 evidence: S 连冷水都能反应, 最活泼; P 气泡多又快; R 少又慢; Q 没有反应

Q3 fair test: same size, same volume, same concentration, same temperature; 可以用 lit splint 证明气体是 hydrogen

Watch for:

Q2 只排顺序不写证据只有 A (L16)

Teaching line, not a question slide.

时间 1:27-1:28。

L10 和 L11 来自她 booklet p.4 的表；L14 来自 p.17 的 tube B；L12 来自 p.6。

L16 在作业 hw2 里再练。

Teaching line, not a question slide.

时间 1:28-1:29。

同样让她先说再看。

Teaching line, not a question slide.

时间 1:29–1:40。

答案在 classwork_answers.pdf: Part D 的 density 题答案是 2.7 g/cm^3 (aluminium); Part E 重点是 alloy 的解释链; Part F 的实验设计要有三个 controlled variables。

1:38 收卷, 准备发 Mini mock。

Teaching line, not a question slide.

时间 1:40–1:57。

发 mock.pdf，计时 17 分钟，老师不提示。

答案和评分标准在 mock_answers.pdf，每题写了 A、M、E 各需要写到哪一步。

时间到了先收卷，课后批改，下次课开头花 5 分钟讲最差的两题。

Teaching line, not a question slide.

时间 1:57-2:00。

hw1: pollination、osmosis、transpiration 的 Explain 题和一个 Spot the error; hw2: density 计算、reactivity 数据题、rusting 实验设计和一个 Spot the error。

答案文件 hw1_answers.pdf 和 hw2_answers.pdf, 同版面红色答案。

booklet p.10 的 solder 图: 提醒 SLurP of Tea, x 轴是 percentage of tin。